

Research Questions and Manuscript Navigation

This paper addresses eleven core research questions, each developed in specific sections and yielding a principal contribution (**C1–C11**, as enumerated in ??). The table below provides a navigational overview; the reading guide that follows explains the logical dependencies between questions.

#	Research Question	Addressing	
		Sec- tions	C
RQ1	Can linguistic case systems—across all major typological alignment patterns—be formalized within a single algebraic framework, with roles as objects and grammatical relations as morphisms?	??, ??	C1
RQ2	Do string diagrams derived from pregroup type reductions provide a computationally and cognitively superior representation of case derivations compared to linear notation?	??, ??, ??	C2
RQ3	Can the DisCoCat meaning functor be extended with case-typed noun spaces to unify formal and distributional semantics—and does this	??, ??, ??, ??	C3